

Q.30 List any four advantages and disadvantages of cycloconverter. (CO3)

Q.31 Explain the function of free wheeling Diode. (CO2)

Q.32 What do you mean by Class a Chopper. Explain? (CO3)

Q.33 Explain Block diagram of Dual converter. (CO3)

Q.34 Differentiate between AC drive and Dc drive. (CO4)

Q.35 Explain in brief Slip control drive. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain construction working principal and VI characteristics of TRIAC with diagram and Waveform. (CO1)

Q.37 Explain Speed control of DC motor using chopper with circuit Diagram. (CO4)

Q.38 What is cyclo converter. Explain single phase to to single phase step up cycloconverter. (CO3)

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6th Sem / Elect.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A DIAC is _____ Device (CO1)

- a) Unidirectional b) Bidirectional
- c) Tri Directional d) None of these

Q.2 A fully controlled converter use (CO2)

- a) Diodes only b) Both A & B
- c) Thyristor Only d) None of these

Q.3 A uni junction transistor has _____ (CO1)

- a) Two PN junction b) One PN junction
- c) Three PN junction d) None of these

Q.4 A SCR is a _____ layer _____ junction device (CO1)

- a) 2,3 b) 4,4
- c) 4,3 d) 3,4

Q.5 Storage Device for UPS (CO5)

- a) Batteries b) SMPS
- c) CVT d) Diode

- Q.6 For the wide range of speed control _____ are used (CO4)
- a) Ac drives b) Dc drives
c) Transformer d) Fuse
- Q.7 Speed of Motor increase with increase in ____ (CO4)
- a) Frequency b) Power factor
c) Armature Voltage d) Slip
- Q.8 With step up chopper, Four Quadrant Operation is possible (CO3)
- a) True b) False
- Q.9 In Dual converter, converter 1&2 works as (CO3)
- a) Rectifier, Rectifier b) Inverter, Rectifier
c) Inverter, Inverter d) None of these
- Q.10 TRIAC, Behaves Like two SCR connected _____ (CO1)
- a) Series b) Parallel
c) Antiparallel d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is Snubber circuit? (CO1)
- Q.12 What is heart of an inverter? (CO3)
- Q.13 What is Freewheeling Diode? (CO2)

- Q.14 Full form of UPS. (CO5)
- Q.15 Define AC drive. (CO4)
- Q.16 Define VFD. (CO4)
- Q.17 Define Holding current. (CO1)
- Q.18 Define Dual converter. (CO3)
- Q.19 What is step down chopper? (CO3)
- Q.20 Draw the symbol of DIAC and list its one application. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Compare SCR and TRIAC. (CO1)
- Q.22 Explain two transistor Analogy of SCR. (CO1)
- Q.23 Explain working of step down chopper. (CO3)
- Q.24 Explain single Phase Half controlled Full wave rectifier. (CO2)
- Q.25 Explain in brief on line UPS. (CO5)
- Q.26 Explain in brief variable Frequency Drive. (CO4)
- Q.27 What kind of Storage Devices used for the UPS. (CO5)
- Q.28 Explain Series Inverter. Write down its important Applications. (CO3)
- Q.29 Explain in brief selection of heat sink for SCR. (CO1)